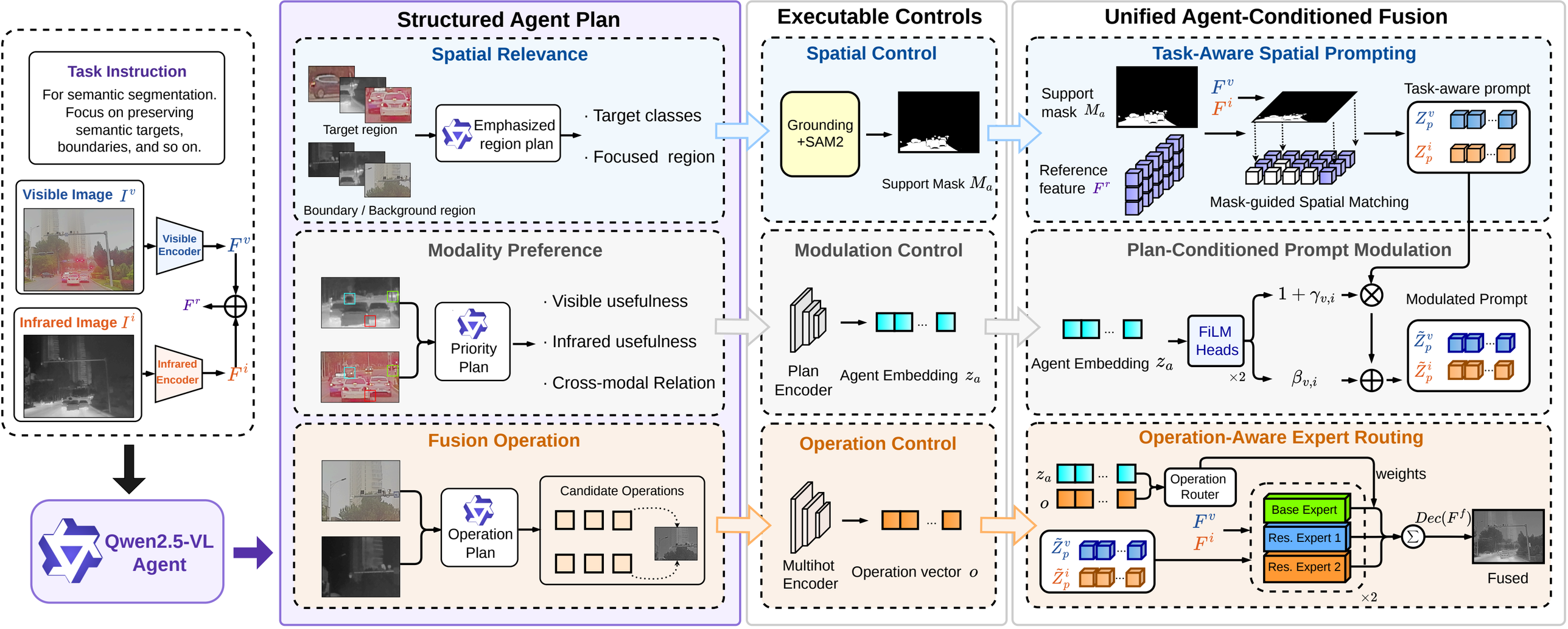

Task Instruction
For semantic segmentation. Focus on preserving semantic targets, boundaries, and so on.
Visible Image I^v
Visible Encoder
F^v
F^r
Infrared Image I^i
Infrared Encoder
F^i
Qwen2.5-VL Agent
Structured Agent Plan
Spatial Relevance
Target region
Boundary / Background region
Emphasized region plan
· Target classes
· Focused region
Modality Preference
Priority Plan
· Visible usefulness
· Infrared usefulness
· Cross-modal Relation
Fusion Operation
Operation Plan
Candidate Operations
Executable Controls
Spatial Control
Grounding +SAM2
Support Mask M_a
Modulation Control
Plan Encoder
Agent Embedding z_a
Operation Control
Multihot Encoder
Operation vector o
Unified Agent-Conditioned Fusion
Task-Aware Spatial Prompting
Support mask M_a
Reference feature F^r
Mask-guided Spatial Matching
Task-aware prompt
Z_p^v
Z_p^i
Plan-Conditioned Prompt Modulation
Agent Embedding z_a
FiLM Heads
×2
$1+\gamma_{v,i}$
$\beta_{v,i}$
Modulated Prompt
$\tilde{Z}_p^v$
$\tilde{Z}_p^i$
Operation-Aware Expert Routing
z_a
o
Operation Router
weights
$\tilde{Z}_p^v$
$\tilde{Z}_p^i$
F^v
F^i
Base Expert
Res. Expert 1
Res. Expert 2
×2
$Dec(F^f)$
Fused